



Systems Design

Week 3: Interconnected Circles
Feb 3, 2026

Pia Zaragoza

Agenda

- 6:30pm - 8:00pm: Discuss Dr. Wachter's Talk / Review Homework
- 8:00pm - 8:10pm: Stretch Break
- 8:10pm - 8:45pm: Lecture
- 8:45pm - 9:20pm: Studio Time

Dr. Bob Wachter

Reflections

- What stood out to you?
- Did anything surprise you?
- Any initial thoughts on potential leverage points or “places to intervene in the system”?



Dr. Bob Wachter

- “We can’t hire radiologists fast enough”.
- Electronic Health Records: “It didn’t transform healthcare and in some ways it made it worst.”
- “In healthcare we innovate, the needs are insatiable and the goals has been to hire more humans. It has been a relentless job creator.”
- I recognize the failure in being that its good at taking in data, but not good at processing it and visualizing it.
- “Human in the loop”
- “Humans suck at remaining vigilant over time”
- “Fail fast doesn’t work because we can’t kill people”.



Patient Journey Map



Healthcare professionals along the patient journey



Treatment / Prognosis



**What is
blood cancer?**

B

Radical Health



Radical

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Get a personalized report explaining your cancer and treatment options.

Share your concerns and we'll analyze your records using the latest research and data

Ask a question



What clinical trials are for me?

Am I on the right treatment?

How do I manage a side effect?

Causal Loop Diagrams

CLD 101

Causal Loop Diagrams (CLD) are great for:

- “Quickly capturing your hypotheses about the causes of dynamics.
- Eliciting and capturing the mental models of individuals and teams.
- Communicating the important feedback processes you believe are responsible for a problem.

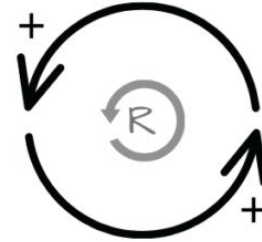


Figure 1.1 from [Müller, FMP, Springer 2015]

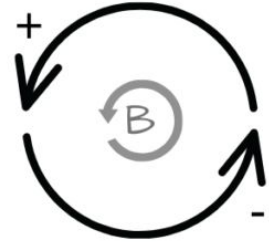
Feedback Loops

There are two kinds of feedback loops: reinforcing and balancing.

- In a **reinforcing loop**, the elements repeatedly perform the same action, resulting in exponential growth or decay.
- In a **balancing loop**, the elements of the system balance each other out.
- Both loops can bring along positive or negative consequences.



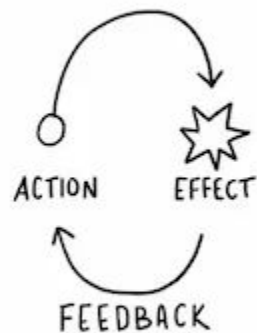
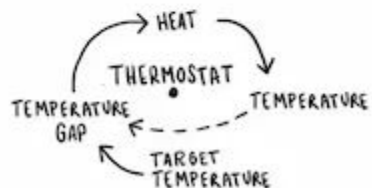
reinforcing
often results in
exponential increase/decrease



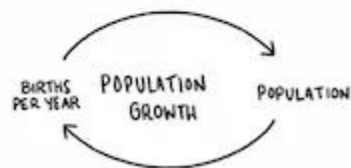
balancing
often results in
reaching a plateau

FEEDBACK LOOPS

- = +
BALANCING
PRODUCES STABILITY
SELF-CORRECTING



+ = -
REINFORCING
LEADS TO INSTABILITY
EXPONENTIAL GROWTH

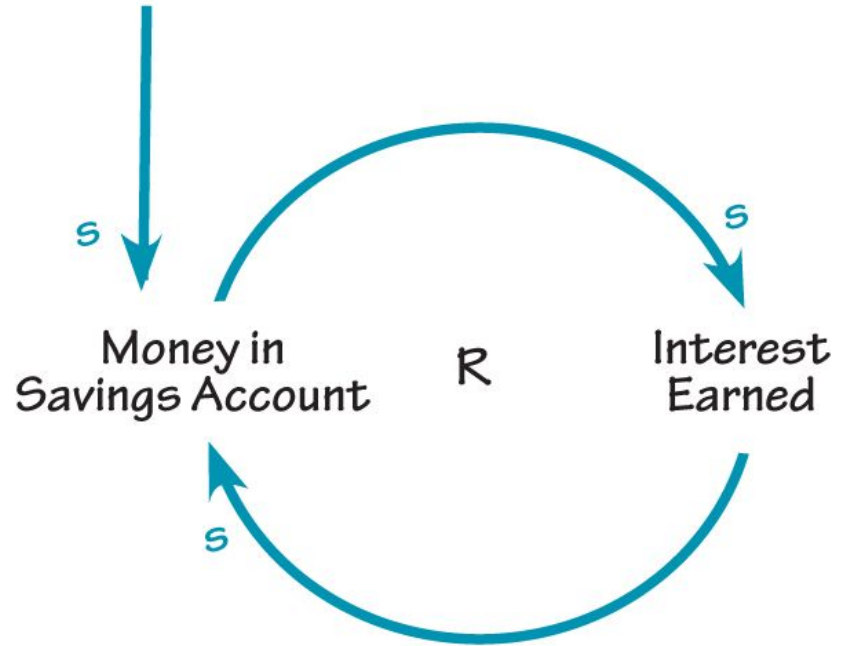


Cheat Sheet

- A **causal loop diagram** consists of four basic elements: the variables, the links between them, the signs on the links (which show how the variables are interconnected), and the sign of the loop (which shows what type of behavior the system will produce).
- The first step in creating a causal “story” is to identify the nouns or variables that are important to the issue. Remember, a variable is something that can vary over time.
- Draw an arrow from the cause to the effect.
- Indicate an increase (draw a "+") or a decrease (draw a "-") for the second element in each relationship.
- Create the loop diagram by adding all the variables and linking them with s (same direction) or o (opposite direction) connections.
- To determine if a loop is reinforcing or balancing, one quick method is to count the number of “o’s.” If there are an even number of “o’s” (or none are present), the loop is reinforcing. If there are an odd number of “o’s,” it is a balancing loop.
- When designing a causal loop diagram, we focus on the accurate illustration of an individual’s beliefs instead of capturing the world as it is.
- Systems mapping helps us gain new insights by organizing experiences and motivations in a relational framework.

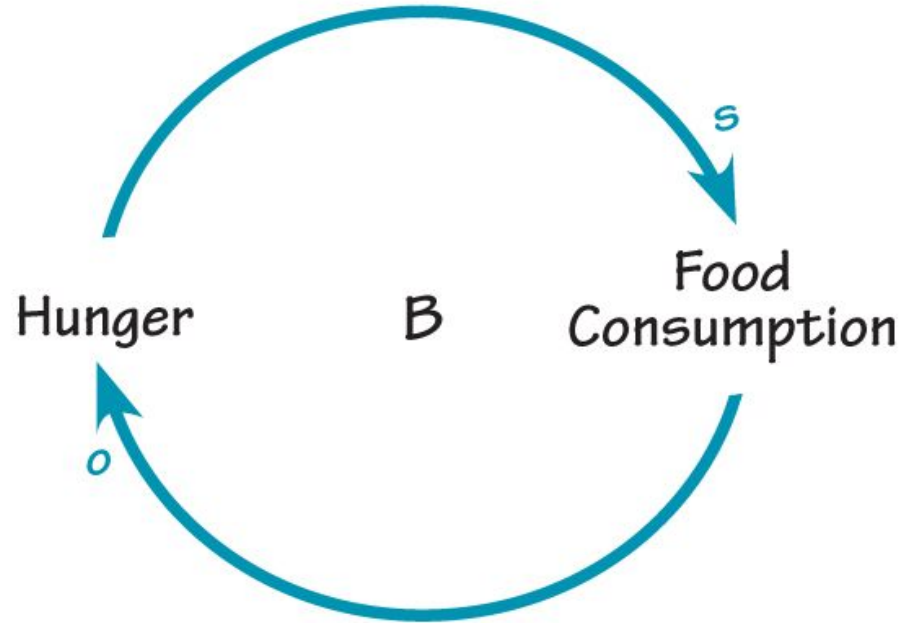
Example: Money

- **Reinforcing loops** change in one direction and is compounded by more change. S means the same direction connections.
- Draw an arrow from the cause to the effect.
- Money in a savings account generates interest, which increases the balance in the savings account and earns more interest.



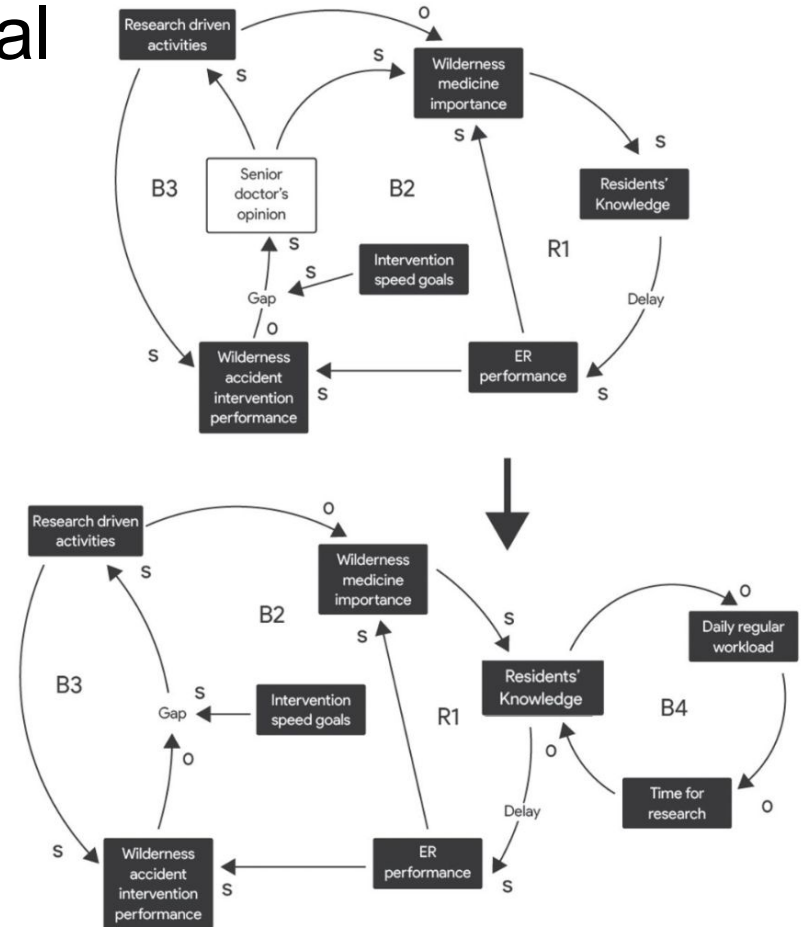
Example: Meals

- **Balancing loops** counter change in one direction with change in the opposite direction.
- Balancing processes attempt to bring things to a desired state and keep them there (thermostat).
- When we are hungry, our body sends a signal to our brain that it's time to eat, which appeases the hunger.



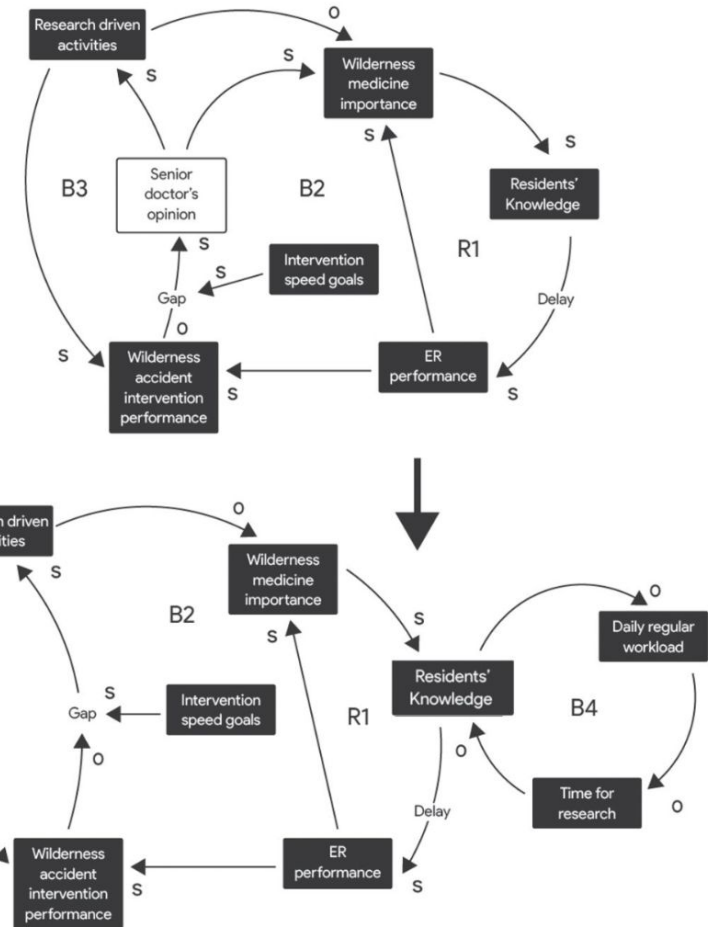
Example: Residents in Hospital

- **Problem:** In one hospital, senior ER doctors asked a group of residents to improve their knowledge about wilderness medicine. The residents formed a wilderness medicine research team. The research team had to read through long documents, previous data on patient handling, and areas that lack proper resources.
- **Key Elements:** Some of the most important elements could be the following: “Senior Doctors’ Expectations,” “Wilderness Medicine Improvement’s Importance,” or “Regular ER Duty Overload.”
- **Relationship between Elements:** Look for causes, consequences, and constraints that bind these variables. We can connect “Too Much Work” with the “Lack of Time and Energy” variable with an S line. This means they change in the same direction, the more ER workload the residents have, the more they will lack the time and energy to do their research work.



Example: Residents in Hospital

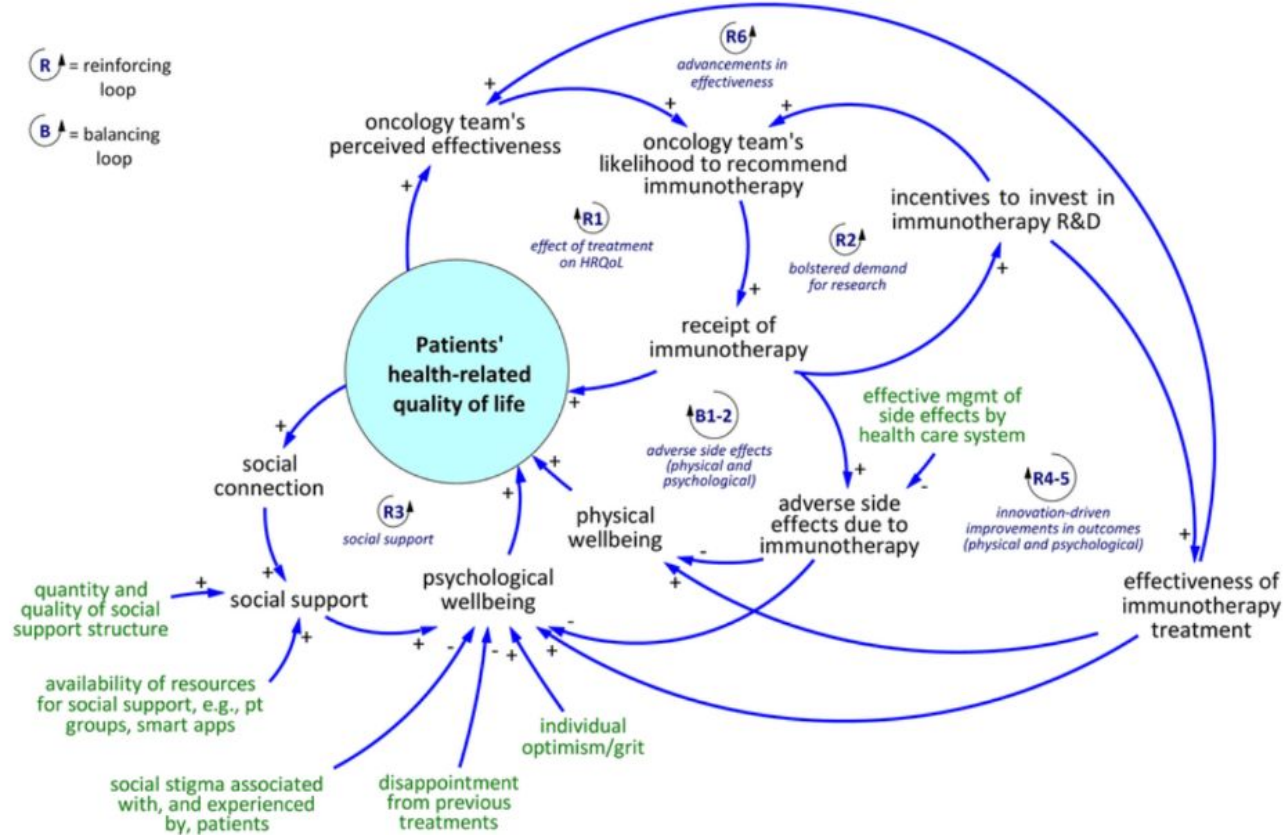
- **Tell the story:** The hospital story's title could be "Improvement without Time and Energy" to portray the resident team's dynamics. The research effort failed because the people who were responsible for it were overworked and tired. For the residents, the researching task was secondary to their more urgent daily demands.
- The first causal loop diagram (top) describes senior doctors' beliefs about the importance of wilderness medicine. Research and development about it would increase wilderness accident-related intervention quality and speed (R1).
- However, they were also focused on research-driven activities that would reduce the gap between wilderness accident intervention performance and the overall ER performance (B2 and B3).
- In the bottom diagram, Loop B4 is added to capture the dynamics of regular work demands that take the residents' time away from research activities. Senior doctors believed that as awareness and skillfulness of wilderness accident intervention increased, the ER doctors' quality of problem handling would also increase. This would eventually lead to improved long-term medical performance, further proving the importance of research and development in specific ER fields (Loop R1).



Instructions

- 1) **Choose the problem:** To get started with your causal loop diagram, you need to find an exact problem in a system you want to investigate. After finding the central problem, we need to collect information about its elements, interconnections, and boundaries.
- 2) **Select key elements and “scrub”:** Once you have the initial story of the problem, it's time to find the most critical elements. These could be identified easily if you prepared a cluster and interconnected circle map. Whichever elements ended up having the most links could be considered key elements.
- 3) **Describe the relationship between elements:** Step three requires you to find other variables that are closely connected to every significant variable. The more interconnections you identify, the better. Look for causes, consequences, and constraints that bind these variables.
- 4) **Tell the story:** We need to find several variable pairs to find a shared story running through them all. The shared story will guide us in developing a model that includes only the problem's relevant aspects. The story should have clear feedback mechanisms, which means the events presented should be recurring.
- 5) **Build a causal loop:** Create the loop diagram by adding all the variables and linking them with s (same direction) or o (opposite direction) connections. Add variables until the full story is told.
- 6) **Finalize the causal loop diagram:** When you're done sketching your diagram, double-check it to make sure it illustrates the story correctly. Track down logical fallacies if there are any. Make sure that all the cause-and-effect relationships are well presented.

Quality of Life in Cancer Immunotherapy



Last Week

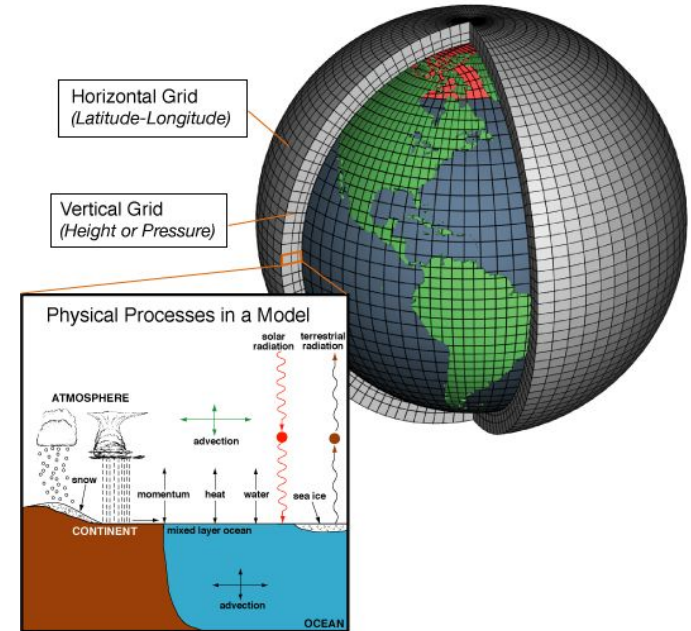
Mental Models

- Mental models are the conscious or unconscious driving force behind our thinking and decision-making.
- Some of the mental models that guide our life we learned from our parents, community, and culture. Some others we adopted by choice. And we rely on them more than we realize when we're trying to **make sense of our surroundings.**
- "Our brain already rehearsed the mental model of traffic lights so frequently that it became an automatic response".



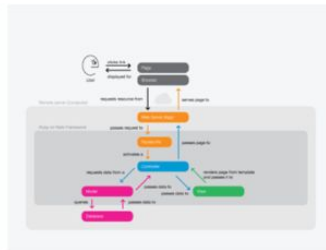
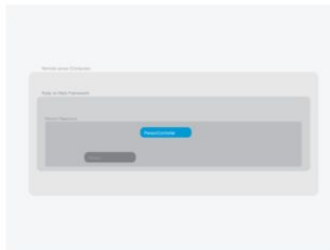
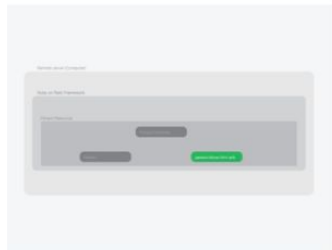
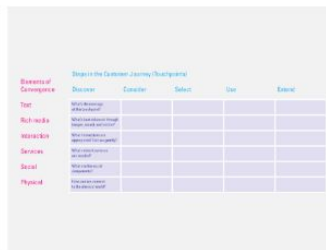
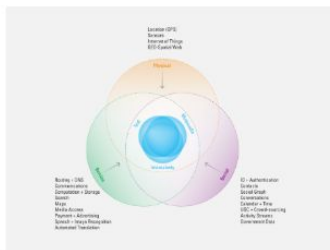
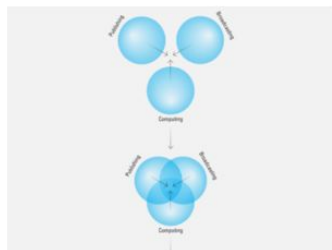
Key Considerations

- Models are **imperfect and incomplete** imitations of the real world.
- A model won't give you a clear yes or no answer to almost any question
- Models are meant to **paint a broader picture**. They are useful in understanding the underlying causes of patterns.
- To become a great modeler, you have to become very good at asking one question. Namely, why?



An example of how the Earth is broken down into cubes displaying some of the many different factors climate modelers consider when creating models. -NOAA

Models



Organizations

Apple

Cooper Design

HfG Ulm

IDEO

Kaiser Permanente

Microsoft

Novell

Starbucks

People

Alan Cooper

Audrey Crane

Austin Henderson

Bill Verplank

Brian Lawson

Chanpory Rith

Charles Eames

Christopher Alex...

Claude Shannon

Claudius Ptolemy

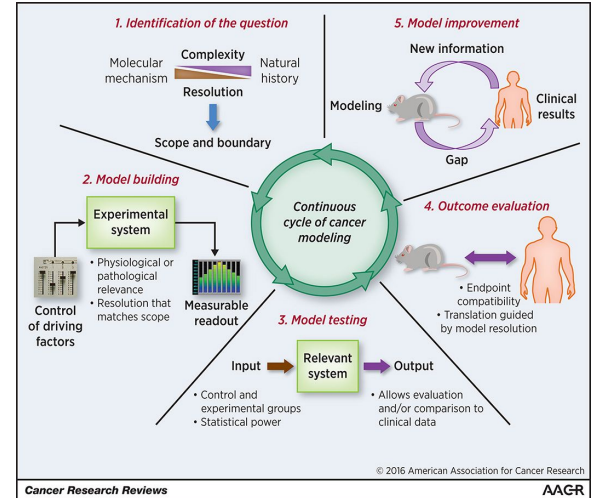
Don Norman

Eric Raymond

Horst Rittel

Define

- One should ask a specific question, a central issue the model is meant to explore. In other words, we need to define the boundaries of the problem we want to model within the system. This defined question will be the guidepost for your model.
- The defining question will lead you to draw boundaries around the elements and interconnections that are important to answer the question and to leave out the parts that are not.
- A model without a clearly defined question becomes directionless, cluttered, and unhelpful.
- A good modeler keeps their assumptions in check and knows that both under- and over-generalizing are detrimental.
- “All models are wrong, but some are useful.” -George E.P. Box



Mental Models

Cognitive maps, mind maps, and concept maps **look and feel similar**; this similarity causes confusion. They are three different ways of visualizing a mental model.

- **Cognitive Maps:** no visual rules that they need to obey.
- **Mind Map** all nodes (except the tree root) have **only one parent node**. Each node can have children corresponding to that concept's subtopics.
- **Concept Map** a graph in which nodes represent concepts and are related through labeled, directed edges that **illustrate relationships between them**.

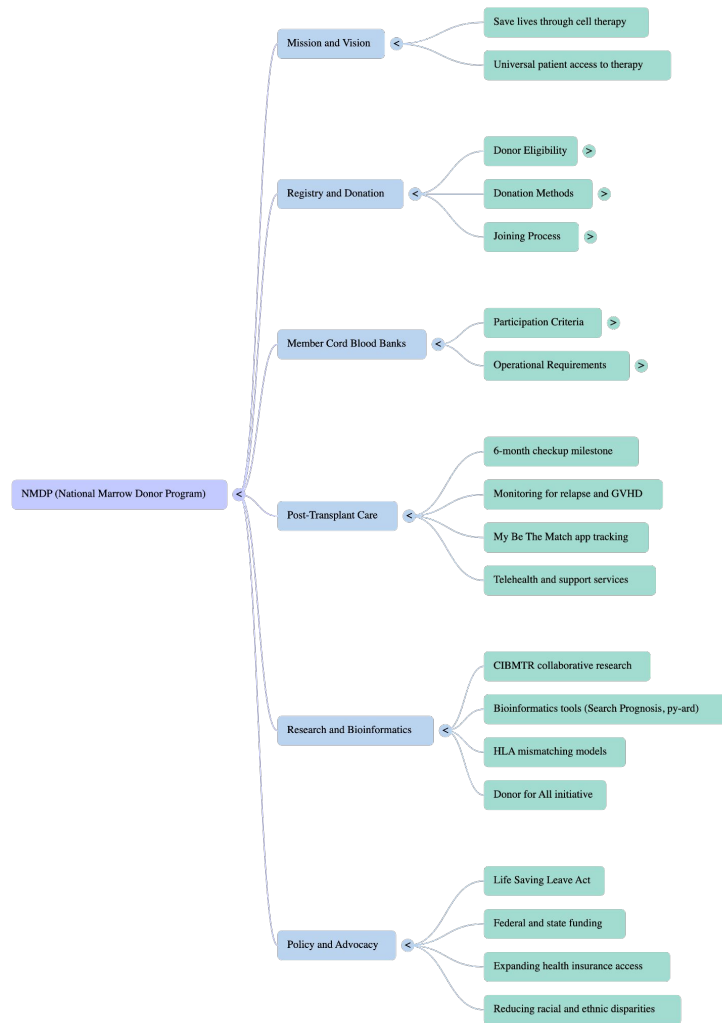


Notebook LLM

- Mind Maps
- Infographics
- Data Table
- Reports
- Slides



Mind Map



Review Homework

Stretch Break

NMDP Deep Dive

Considerations

- **Technology:** What is their technology stack? AI? Databases? Matching software proprietary or off the shelf? Health Insurance Portability and Accountability Act (HIPAA) compliant?
- **Operations:** What are their products, programs, and services? Who do they interact with (e.g non-profits, registries, labs, couriers, and hospitals)?
- **Policy:** How do policy, regulations, and governance shape what is possible in the system?
- **Economics:** What are the funding sources of the NMDP? Any short-term vs. long term risks? Are the incentives misaligned or aligned for patients, donors, employees, ect.?
- **Trust & Power:** What factors affect trust, if any? What are the power dynamics?
- **Diversity, Equity, and Inclusion:** What are the systemic issues? Historical harms?
- **Education:** What are the knowledge gaps or misunderstandings that influence patient or donor decision-making?
- **Failure States:** What happens when the system breaks? What are the other ways to find a donor match?



"If I could tell my donor something, I would probably tell them thank you. Thank you for donating so that I could keep having Christmases, Thanksgivings, birthdays, fun and just being alive."

- Hailey, transplant recipient

TWO TRANSPLANTS, ONE AMAZING GIRL

In February 2020, Hailey began experiencing unusual symptoms. Following alarming blood test results, more testing concluded that she had aplastic anemia.

Hailey was just 7 years old at the time of her diagnosis. Five months later, she underwent a blood stem cell transplant and afterwards struggled with complications of pneumonia and a serious staph infection. With the ongoing threat of the pandemic, Hailey and her family spent a year in isolation to protect her fragile health. Despite the challenges, she eventually returned to school, gradually getting back into the activities she loved, like competitive dance.

In May 2023, Hailey's health began to decline, leading to a heart-breaking diagnosis of leukemia. After intensive treatment with multiple rounds of chemotherapy, another blood stem cell transplant was her best chance for a cure. Hailey received her second transplant in November 2023, one day before her 11th birthday.

It's been over a year since then, and Hailey continues to get stronger—giving her even more reason to dance.

When Hailey was hurting, help was there

To care for her young daughter, Hailey's mom took an unpaid leave of absence, leaving their six-person family to rely on one income for several months. But NMDP—and you—were there.

Thanks to our supporters, Hailey's family was able to receive one of our financial assistance grants. This allowed her mom to better focus her attention on what mattered most—Hailey's recovery.

In 2024, NMDP provided \$6.6 million in patient assistance to 3,300 families.

Hailey's dance dreams come true

For the third season, NMDP was honored to be the beneficiary of the **Minnesota Vikings Field Goals for Charity**. For every successful Vikings field goal, Innovative Office Solutions and Post-it® Brand generously contributed essential funds to support NMDP's mission. This exciting initiative also brought joy to Hailey, who was able to be a part of the on-field festivities.

Despite the challenges life has presented, Hailey's passion for dance has been unwavering. She was overjoyed by the opportunity to meet, practice and appear on the field with the Minnesota Vikings cheerleaders.



FY2024 AT A GLANCE*

Big numbers. Big dreams.

What you see below are more than just impressive numbers; each one represents the support you give, the money you donate and the work you do, all in the name of making dreams possible.

185+
RESEARCH STUDIES
and clinical trials

7,994
LIVES IMPACTED
through cell therapy

\$6.6M
IN ASSISTANCE
provided by NMDP and
our philanthropic partners



LIVES

- **7,994** patient lives impacted through cell therapy—more than any other year since our founding in 1987
- **140,000+** lives impacted through cell therapy since 1987

RESEARCH

- **~\$12.5 million** invested since 1998 to fund **51 scholars** with the NMDP Amy Strelzer Manasevit Research Program
- **185+** ongoing research studies and clinical trials through CIBMTR® (Center for International Blood and Marrow Transplant Research®), a research collaboration between the Medical College of Wisconsin® and NMDP
- **310+** transplant centers participating in research

RECRUITMENT

- **240,000+** potential donors joined the NMDP Registry™
- **85,000+** ethnically diverse potential donors joined the NMDP Registry
- **42 million+** potential donors available for every patient search
- **180,000+** potential donors on the NMDP Registry in Mexico

PATIENT ASSISTANCE

- **3,300** patient families received NMDP grants to financially support them through treatment
- **\$6.6 million** in patient assistance provided by NMDP and our philanthropic partners

MISSION SUPPORT

- **17,000+** individual contributors made financial gifts to NMDP
- **\$19 million+** raised by NMDP supporters and contributors
- **\$7 million** worth of hours donated by volunteers

“Being called was like winning the lottery. I was handed a life-changing chance to save a life and there was no way I was saying no.”

– Libby, blood stem cell donor

“NMDP is the perfect intersection of science and love.”

– Bill (NMDP Foundation board member), father of Riley, transplant recipient

NATALIE, BLOOD
STEM CELL DONOR



“It was really important
to have the wage
reimbursement benefit for
me to donate with peace of
mind. I am self-employed
and wouldn't have been able
to receive paid time off.”

- Karina, blood stem cell donor

ADVANCING OUR MISSION

Donor financial assistance

Becoming a hero as a blood stem cell donor requires a significant time commitment. For those who don't have wage coverage benefits, NMDP typically provides reimbursement for lost wages and other financial support.

In 2024, NMDP donor financial assistance supported 447 donors and 153 donor companions, marking a 17% increase from the previous year.

Natalie joined the registry as a college student and, years later, received the call that she was a match for a patient in need. She quickly said “yes,” and within weeks, she and her companion were on a three-hour flight to help save a life. Forty percent of donors have to travel to complete their donation.

Patient financial assistance

Financial concerns shouldn't keep patients from accessing potentially life-saving cell therapy—and making dreams possible. Thanks to our financial supporters, NMDP offers grants to help patients like Liam and their families.

Liam was born with Fanconi anemia, a rare genetic disorder affecting his bone marrow. His best chance for a cure was a blood stem cell transplant. His family moved from Colorado to Minnesota for the transplant, facing challenges such as costly longer-term housing and the need to live on one income.

Thanks to your support, we provided Liam's family with financial assistance, allowing them to focus on Liam's health rather than financial stress. Because with NMDP, **good goes a long way.**

IN 2024, WE PROVIDED \$6.6 MILLION IN FINANCIAL ASSISTANCE GRANTS TO 3,300 FAMILIES.

“Seeing Liam being able to be a regular kid has been the biggest blessing. The compassion each of you encompasses by supporting an incredibly important cause is immeasurable.”

- Xandra, Liam's mom



Liam with his mom, Xandra



ADVOCACY IN ACTION

NMDP is committed to saving lives by advancing public policies that support our mission.

2024 achievements

PUBLIC INVESTMENTS

Secured \$33 million for the C.W. Bill Young Cell Transplantation Program, matching the largest federal contribution ever.

ACCELERATING KIDS' ACCESS TO CARE ACT

Mobilized thousands of volunteers to support the act, which passed the U.S. House, simplifying out-of-state care for children receiving Medicaid.

EMPOWERING STUDENT DONORS

Developed an advocacy toolkit to help college chapters update attendance policies for student donors.

STATE POLICY WINS

Partnered with patient advocacy groups to support 13 pieces of legislation, 11 of which became laws focused on growing the national registry, enhancing support for caregivers and more.

VOLUNTEER ADVOCATE PROGRAM

Worked with advocates to build relationships with many members of Congress.

Attendees of
The ONE Forum by NMDP



FINANCIALS

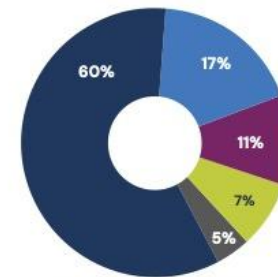
NMDP and subsidiaries consolidated statement of financial position as of Sept. 30, 2024 (in thousands)

Assets	
Current and other assets	\$163,931
Long-term investments	\$152,923
Property and equipment	\$79,593
TOTAL	\$396,447
Liabilities and net assets	
Current and long-term liabilities	\$169,960
Net assets	\$226,487
TOTAL	\$396,447

Consolidated statement of activities for the year ending Sept. 30, 2024 (in thousands)

Revenues and gains	
Search and procurement fees	\$560,350
Federal contracts, cooperative agreements and contributions	\$85,828
Investment and other income	\$200
TOTAL	\$646,378
Expenses	
Program services	\$529,304
Management and general	\$104,815
Fundraising	\$4,716
TOTAL	\$638,835
Non-operating income	\$29,209
INCREASE IN NET ASSETS	\$36,752

HOW WE SPEND EACH DOLLAR



Medical Services

Management & Fundraising

Research

Recruitment

Public Awareness

Scan to read our consolidated audited financial statements with supplemental schedules



National Marrow Donor Program and Subsidiaries

Consolidated Statement of Functional Expenses

Year Ended September 30, 2024

(In Thousands)

	Program Services					Support Services			Total
	Medical Services	Recruitment	Research	Public Awareness	Total	Management and General	Fundraising	Total	
Medical services	\$ 247,335	\$ -	\$ -	\$ -	\$ 247,335	\$ 2,376	\$ -	\$ 2,376	\$ 249,711
Donor recruitment services	-	12,752	-	-	12,752	-	-	-	12,752
Research activities	-	-	15,901	-	15,901	-	-	-	15,901
Compensation	82,497	17,716	25,620	13,502	139,335	48,307	1,555	49,862	189,197
Benefits	21,287	4,530	7,222	3,502	36,541	11,118	477	11,595	48,136
Professional fees	13,593	6,384	14,652	8,694	43,323	15,567	1,997	17,564	60,887
Travel	1,086	1,378	735	570	3,769	1,261	124	1,385	5,154
Professional development	578	126	130	92	926	585	8	593	1,519
Occupancy	379	94	24	22	519	3,928	-	3,928	4,447
Telecommunications	478	117	115	115	825	357	-	357	1,182
Information processing	9,152	2,343	3,033	2,422	16,950	7,515	8	7,523	24,473
Printing and copying	145	164	25	526	860	445	38	483	1,343
Postage and shipping	405	190	56	159	810	124	22	146	956
Depreciation and amortization	4,056	971	939	931	6,897	8,541	-	8,541	15,438
Interest expense	398	54	74	27	553	1,787	-	1,787	2,340
Office and miscellaneous	525	275	67	1,141	2,008	2,904	487	3,391	5,399
Total expenses	\$ 381,914	\$ 47,094	\$ 68,593	\$ 31,703	\$ 529,304	\$ 104,815	\$ 4,716	\$ 109,531	\$ 638,835

See notes to consolidated financial statements

National Marrow Donor Program and Subsidiaries

Consolidated Statement of Functional Expenses

Year Ended September 30, 2023

(In Thousands)

	Program Services					Support Services			Total
	Medical Services	Recruitment	Research	Public Awareness	Total	Management and General	Fundraising	Total	
Medical services	\$ 222,531	\$ -	\$ -	\$ -	\$ 222,531	\$ 1,037	\$ -	\$ 1,037	\$ 223,568
Donor recruitment services	-	15,252	-	-	15,252	-	-	-	15,252
Research activities	-	-	14,141	-	14,141	-	-	-	14,141
Compensation	73,231	16,423	21,785	12,745	124,184	47,553	2,419	49,972	174,156
Benefits	18,921	3,922	5,325	3,218	31,386	11,345	607	11,952	43,338
Professional fees	14,397	6,776	16,857	4,098	42,128	14,232	1,557	15,789	57,917
Travel	887	1,265	514	686	3,352	1,281	18	1,299	4,651
Professional development	527	75	113	81	796	508	4	512	1,308
Occupancy	489	93	45	144	771	3,898	51	3,949	4,720
Telecommunications	522	130	127	64	843	395	63	449	1,292
Information processing	7,427	1,869	2,869	977	13,142	6,278	902	7,180	20,322
Printing and copying	154	90	47	391	682	135	122	257	939
Postage and shipping	404	187	38	119	748	237	15	252	1,000
Depreciation and amortization	5,166	1,316	1,234	677	8,393	8,808	577	9,385	17,778
Interest expense	-	-	-	-	-	1,703	-	1,703	1,703
Office and miscellaneous	547	279	98	202	1,126	2,088	211	2,299	3,425
Total expenses	\$ 345,203	\$ 47,677	\$ 63,193	\$ 23,402	\$ 479,475	\$ 99,489	\$ 6,546	\$ 106,035	\$ 585,510

See notes to consolidated financial statements

Public Inspection Copy

Form **990**

Return of Organization Exempt From Income Tax

OMB No. 1545-0047

Under section 501(c), 527, or 4947(a)(1) of the Internal Revenue Code (except private foundations)

Department of the Treasury
Internal Revenue ServiceDo not enter social security numbers on this form as it may be made public.
Go to www.irs.gov/Form990 for instructions and the latest information.**2023**Open to Public
Inspection

A For the 2023 calendar year, or tax year beginning OCT 1, 2023 and ending SEP 30, 2024	
B Check if applicable: Address change Name change Initial return Final return/terminated Amended return Application pending	C Name of organization XMDP FOUNDATION Doing business as Number and street (or P.O. box if mail is not delivered to street address) Room/suite 500 N 5TH ST City or town, state or province, country, and ZIP or foreign postal code MINNEAPOLIS, MN 55401-1206 F Name and address of principal officer: ERICA JENSEN SAKE AS C ABOVE
D Employer identification number 41-1704734	
E Telephone number 763-406-8700	
G Gross receipts \$ 34,289,320	
H(a) Is this a group return for subordinates? Yes ☒ No ☐	
H(b) Are all subordinates included? Yes ☐ No ☐ If "No," attach a list. See instructions	
H(c) Group exemption number	
I Tax-exempt status: ☒ 501(c)(3) 501(c) () (insert no.) 4947(a)(1) or 527	
J Website: WWW.XMDP.ORG	
K Form of organization: ☒ Corporation ☐ Trust ☐ Association ☐ Other	
L Year of formation: 1991 M State of legal domicile: MN	

Part I Summary

1 Briefly describe the organization's mission or most significant activities: SEE SCHEDULE O			
2 Check this box ☐ if the organization discontinued its operations or disposed of more than 25% of its net assets.			
3 Number of voting members of the governing body (Part VI, line 1a)	3	17	
4 Number of independent voting members of the governing body (Part VI, line 1b)	4	16	
5 Total number of individuals employed in calendar year 2023 (Part V, line 2a)	5	45	
6 Total number of volunteers (estimate if necessary)	6	3000	
7a Total unrelated business revenue from Part VIII, column (C), line 12	7a	0	
7b Net unrelated business taxable income from Form 990-T, Part I, line 11	7b	0	
Revenue	Prior Year	Current Year	
8 Contributions and grants (Part VIII, line 1h)	18,893,262	22,907,977	
9 Program service revenue (Part VIII, line 2g)	0	0	
10 Investment income (Part VIII, column (A), lines 3, 4, and 7d)	350,756	2,335,205	
11 Other revenue (Part VIII, column (A), lines 5, 6d, 8c, 9c, 10c, and 11e)	-555,030	-938,540	
12 Total revenue - add lines 8 through 11 (must equal Part VIII, column (A), line 12)	18,688,988	24,304,642	
13 Grants and similar amounts paid (Part IX, column (A), lines 1-3)	9,700,536	12,389,221	
14 Benefits paid to or for members (Part IX, column (A), line 4)	0	0	
15 Salaries, other compensation, employee benefits (Part IX, column (A), lines 5-10)	5,791,779	7,240,574	
16a Professional fundraising fees (Part IX, column (A), line 11e)	612,029	482,576	
b Total fundraising expenses (Part IX, column (D), line 25)	2,175,120		
17 Other expenses (Part IX, column (A), lines 11a-11d, 11f-24e)	2,157,082	1,952,631	
18 Total expenses. Add lines 13-17 (must equal Part IX, column (A), line 25)	18,261,426	23,065,002	
19 Revenue less expenses. Subtract line 18 from line 12	427,562	2,239,640	
Net Assets or Fund Balances	Beginning of Current Year	End of Year	
20 Total assets (Part X, line 16)	15,913,956	18,372,366	
21 Total liabilities (Part X, line 26)	1,878,244	1,852,736	
22 Net assets or fund balances. Subtract line 21 from line 20	14,035,712	16,519,630	

Part II Signature Block

Under penalties of perjury, I declare that I have examined this return, including accompanying schedules and statements, and to the best of my knowledge and belief, it is true, correct, and complete. Declaration of preparer (other than officer) is based on all information of which preparer has any knowledge.

Sign Here	Signature of officer KATHERINE HEYERDAHL, CHIEF FINANCIAL OFFICER		Date 08/14/25	
Print/Type preparer's name	KAREN A. GRIES	Preparer's signature	Date	Check if self-prepared
Preparer Use Only	Firm's name BAKER TILLY ADVISORY GROUP, LP	Firm's EIN 39-0859910	Phone no. 612.876.4500	PTIN 900078514

May the IRS discuss this return with the preparer shown above? See instructions

LHA For Paperwork Reduction Act Notice, see the separate instructions.

332001 12-21-25

Form **990** (2023)

Part VII Compensation of Officers, Directors, Trustees, Key Employees, Highest Compensated Employees, and Independent ContractorsCheck if Schedule O contains a response or note to any line in this Part VII ☐**Section A. Officers, Directors, Trustees, Key Employees, and Highest Compensated Employees****1a** Complete this table for all persons required to be listed. Report compensation for the calendar year ending with or within the organization's tax year.

• List all of the organization's **current** officers, directors, trustees (whether individuals or organizations), regardless of amount of compensation. Enter -0- in columns (D), (E), and (F) if no compensation was paid.

• List all of the organization's **current** key employees, if any. See the instructions for definition of "key employee."

• List the organization's five **current** highest compensated employees (other than an officer, director, trustee, or key employee) who received reportable compensation (box 5 of Form W-2, box 6 of Form 1099-MISC, and/or box 1 of Form 1099-NEC) of more than \$100,000 from the organization and any related organizations.

• List all of the organization's **former** officers, key employees, and highest compensated employees who received more than \$100,000 of reportable compensation from the organization and any related organizations.

• List all of the organization's **former** directors or trustees that received, in the capacity as a former director or trustee of the organization, more than \$10,000 of reportable compensation from the organization and any related organizations.

See the instructions for the order in which to list the persons above.

☐ Check this box if neither the organization nor any related organization compensated any current officer, director, or trustee.

(A) Name and title	(B) Average hours per week (list any hours for related organizations below line)	(C) Position (do not check more than one box, unless person is both an officer and a director/trustee)					(D) Reportable compensation from the organization (W-2/1099-MISC/ 1099-NEC)	(E) Reportable compensation from related organizations (W-2/1099-MISC/ 1099-NEC)	(F) Estimated amount of other compensation from the organization and related organizations
		Individual officer or trustee	Officer	Director	Key employee	Highest compensated employee			
(1) AMY RONNESBERG	2.00								
NMDP CHIEF EXECUTIVE OFFICER	48.00	X	X				0.	1,538,663.	455,038.
(2) JOY KING	20.00								
CHIEF ADVANCEMENT OFFICER	30.00			X			230,187.	345,282.	168,631.
(3) KRISTA DUSIL	1.00								
CHIEF FINANCIAL OFFICER (THRU 2/24)	49.00		X				0.	649,755.	53,242.
(4) TODD PETERSON	50.00								
VICE PRESIDENT, ADVANCEMENT	0.00				X		291,111.	0.	84,039.
(5) ANOIE FITZGERALD	50.00								
VP, CORPORATE & FOUNDATION PARTNERSHIPS	0.00			X			274,522.	0.	64,855.
(6) ALEX MENSING	50.00								
SR. DIRECTOR, SUPPORTER EXPERIENCE & ENGAGEMENT	0.00			X			199,924.	0.	46,470.
(7) GINA GRAVES	0.00								
FORMER OFFICER	0.00					X	0.	239,958.	3,353.
(8) KEVIN HEALY	50.00								
DIRECTOR, ADVANCEMENT SERVICES	0.00			X			182,633.	0.	42,949.
(9) DANIEL LEE	50.00								
DIRECTOR, MAJOR AND PLANNED GIFTS	0.00			X			190,612.	0.	32,174.
(10) KATHERINE HEYERDAHL	2.00								
CHIEF FINANCIAL OFFICER (BEGAN 4/24)	48.00		X				0.	0.	0.
(11) MICHAEL STEWART	1.00								
CHAIR	0.00	X	X				0.	0.	0.
(12) MELISSA KONG	1.00								
VICE CHAIR	0.00	X	X				0.	0.	0.
(13) RAMESH SUBRAMANIAN	1.00								
SECRETARY	0.00	X	X				0.	0.	0.
(14) DIANNA CERVANTES	1.00								
TREASURER	0.00	X	X				0.	0.	0.
(15) JASON AHLGREN	1.00								
DIRECTOR	0.00	X					0.	0.	0.
(16) GUSTAVO ALCOCER	1.00								
DIRECTOR	0.00	X					0.	0.	0.
(17) SARAH ABMA	1.00								
DIRECTOR	0.00	X					0.	0.	0.

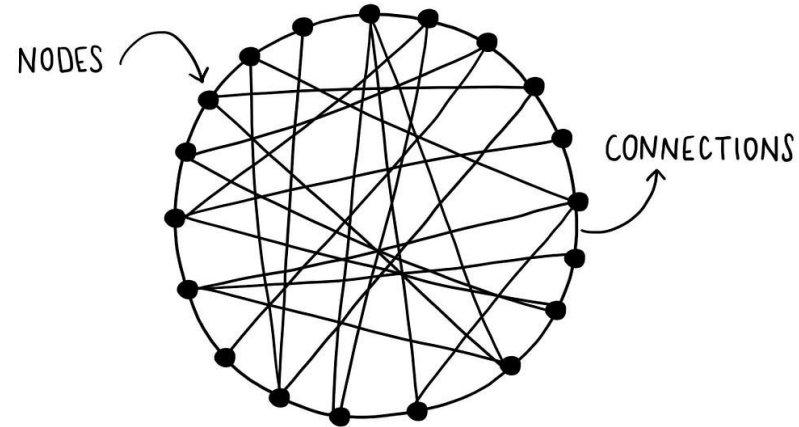
Part VIII Statement of RevenueCheck if Schedule O contains a response or note to any line in this Part VIII ☐

		(A) Total revenue	(B) Related or exempt function revenue	(C) Unrelated business revenue	(D) Revenue excluded from tax under sections 512-514
Contributions, Gifts, Grants, and Other Similar Amounts	1 a Federated campaigns				
	1 b Membership dues				
	1 c Fundraising events	1,649,926.			
	1 d Related organizations	7,586,216.			
	1 e Government grants (contributions)				
	1 f All other contributions, gifts, grants, and similar amounts not included above	13,671,835.			
	1 g Noncash contributions included in lines 1a-1f	\$ 255,801.			
	1 h Total. Add lines 1a-1f	22,907,977.			
	Business Code				
Program Service Revenue	2 a				
	2 b				
	2 c				
	2 d				
	2 e				
	2 f All other program service revenue				
	2 g Total. Add lines 2a-2f				
Other Revenue	3 Investment income (including dividends, interest, and other similar amounts)	463,724.			463,724.
	4 Income from investment of tax-exempt bond proceeds				
	5 Royalties				
		(i) Real (ii) Personal			
	6 a Gross rents	6a			
	6 b Less: rental expenses	6b			
	6 c Rental income or (loss)	6c			
	6 d Net rental income or (loss)				
	7 a Gross amount from sales of assets other than inventory	7a	10,709,627.		
	7 b Less: cost or other basis and sales expenses	7b	8,838,146.		
	7 c Gain or (loss)	7c	1,871,481.		
	7 d Net gain or (loss)		1,871,481.		1,871,481.
	8 a Gross income from fundraising events (not including \$ 1,649,926. of contributions reported on line 1c). See Part IV, line 18	8a	162,044.		
	8 b Less: direct expenses	8b	1,140,588.		
	8 c Net income or (loss) from fundraising events		-978,544.		-978,544.
	9 a Gross income from gaming activities. See Part IV, line 19	9a	45,940.		
	9 b Less: direct expenses	9b	5,944.		
	9 c Net income or (loss) from gaming activities		39,996.		39,996.
	10 a Gross sales of inventory, less returns and allowances	10a			
	10 b Less: cost of goods sold	10b			
10 c Net income or (loss) from sales of inventory					
Business Code					
Miscellaneous Revenue	11 a FOREIGN CURRENCY GAIN	900099	8.		8.
	11 b				
	11 c				
	11 d All other revenue				
	11 e Total. Add lines 11a-11d		8.		
12 Total revenue. See instructions		24,304,642.	0.	0.	1,396,665.

Homework

Interconnected Circles Map

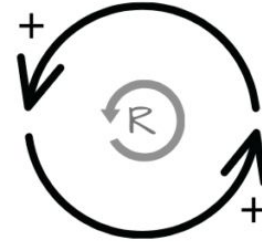
- 1) Start with a circle on a piece of paper.
- 2) Next, identify the key elements of the system you're examining.
How do you know what's a key element? It meets these criteria:
 - It's important to changes in the system
 - Increases or decreases in the system
 - Can be described by a noun
- 3) Write these elements around the circle (don't include more than 10).
- 4) Look for cause and effect.
 - Which elements are directly causing other elements to increase or decrease? Draw an arrow between these elements.
 - Indicate an increase (draw a "+") or a decrease (draw a "-") for the second element in each relationship.
- 5) Find all of the cause-and-effect relationships. These can be based on data or they can be a hypothesis.
- 6) Look for elements whose relationships form closed loops. These are feedback loops.



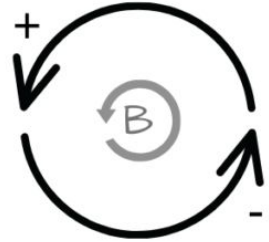
Feedback Loops

There are two kinds of feedback loops: reinforcing and balancing.

- In a **reinforcing loop**, the elements repeatedly perform the same action, resulting in exponential growth or decay.
- In a **balancing loop**, the elements of the system balance each other out.
- Both loops can bring along positive or negative consequences.



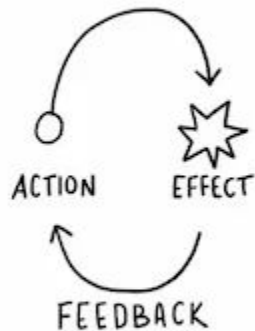
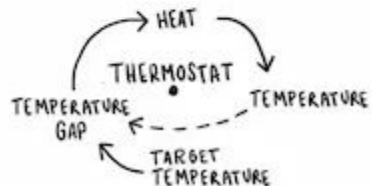
reinforcing
often results in
exponential increase/decrease



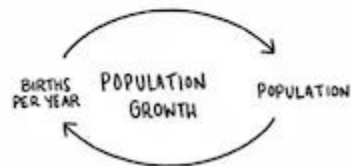
balancing
often results in
reaching a plateau

FEEDBACK LOOPS

- = +
BALANCING
PRODUCES STABILITY
SELF-CORRECTING



+ = -
REINFORCING
LEADS TO INSTABILITY
EXPONENTIAL GROWTH

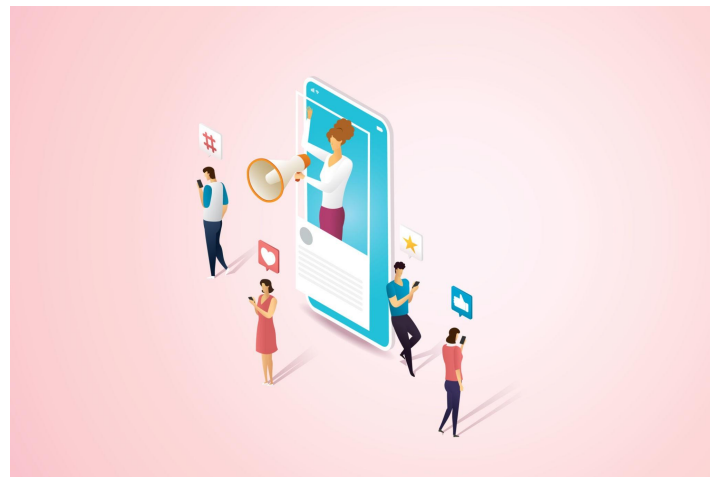


Example

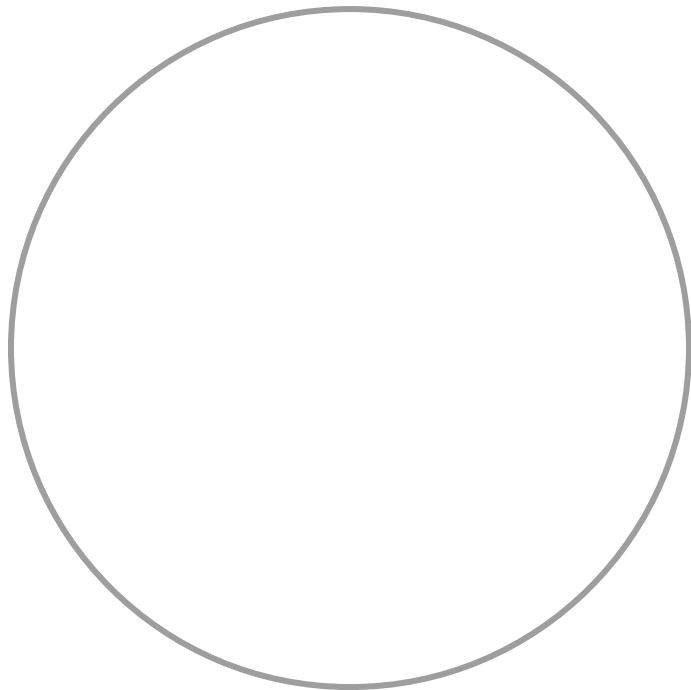
Problem: Our product has been dealing with unhappy customers. We know they've been complaining about too many bugs in the product and slow response times from our support (as customers create more support tickets). At the same time, we've been shipping more features to make them happier.

Key elements: unhappy customers, bugs in the product, response times, support tickets and new features.

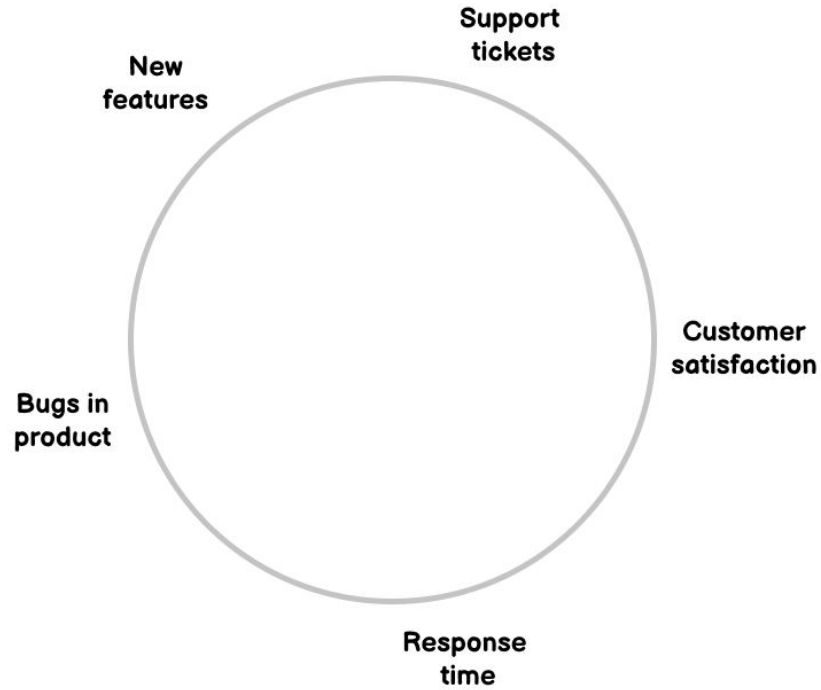
Source: [Untools](#)



Unhappy Customers

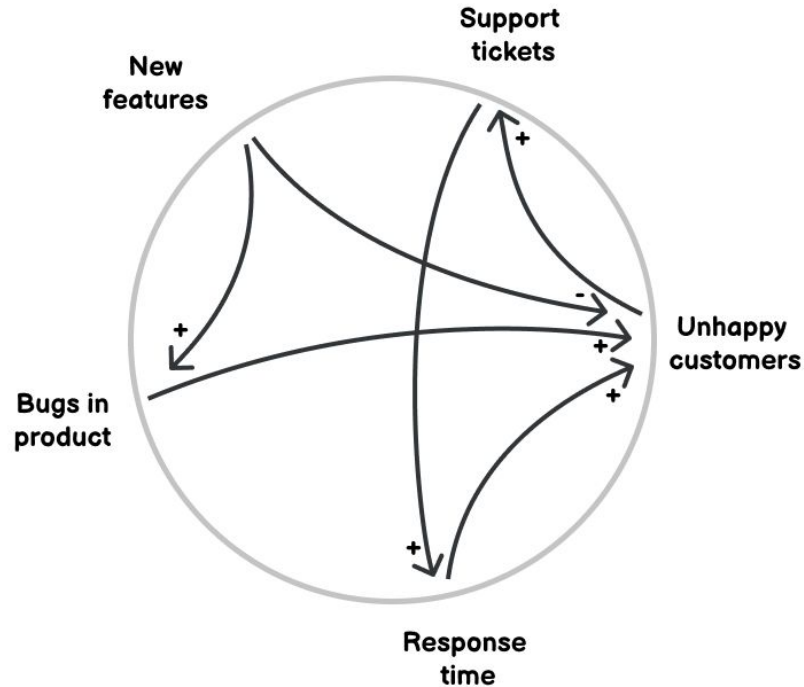


Unhappy Customers



Unhappy Customers

- Draw an arrow from the cause to the effect.
- Indicate an increase (draw a "+") or a decrease (draw a "-").



Thank you!
